Class – 09

COVARIANCE AND CORRELATION

Data Distribution

Covariance

Till now, we were looking at one variable at a time (like marks, height, weight).

But what if we want to know if **two variables are related**?

That's where **Covariance** comes in.

So what does Covariance do??

→ Covariance tells us whether two variables **move together** or **move in opposite directions**.

Covariance

The formula for Covariance is,

$$\mathrm{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n}$$

X_i, Y_i = Data Values

$\bar{X}$, $\bar{Y}$ = Means of X and Y

n = No. of Data Points

Covariance

Now there are two types of Covariance -

- Positive covariance ⟶ When one variable increases the other variable tends to increase

Let's say we look at students' **hours studied (X)** and their **marks (Y)**.

INCREASING ↑

Hours Studied	Marks
2	30
3	35
5	49
7	50
8	65
11	70

INCREASING ↑

Covariance

- Negative covariance ⟶ When one variable increases the other variable decreases.

Let's say we look at students' **Rank (X)** and their **marks (Y)**.

INCREASING

Rank	Marks
2	90
3	80
5	70
7	60
8	65
11	50

DECREASING

Covariance

Let's see an example for Covariance,

Let's say we look at students' **hours studied (X)** and their **marks (Y)**.

Hours Studied	Marks
2	50
4	60
6	65
8	80

$$\text{Mean of X} = \frac{(2+4+6+8)}{4} = 5$$

$$\text{Mean of Y} = \frac{(50+60+65+80)}{4} = 63.75$$

Covariance

Now, for each pair $(X_i - \bar{X})(Y_i - \bar{Y})$

- $(2-5)(50-63.75) = (-3)(-13.75) = 41.25$
- $(4-5)(60-63.75) = (-1)(-3.75) = 3.75$
- $(6-5)(65-63.75) = (1)(1.25) = 1.25$
- $(8-5)(80-63.75) = (3)(16.25) = 48.75$

Sum = 95

Cov(X, Y) = 95 / 4 = **23.75 (positive)**

Correlation

So we learned how covariance told us whether two variables move together?

The problem is, covariance can give **any value** (big or small), so it's hard to compare.

That's where **Correlation** comes in.

→ Correlation tells us **how strong** and **in what direction** two variables are related.

→ It's basically a **scaled version of covariance**, always between **−1 and +1**.

Correlation

The Formula of Correlation is,

$$r = \frac{Cov(X, Y)}{\sigma X \cdot \sigma Y}$$

- Cov(X,Y) = Covariance of X and Y
- σX = Standard Deviation of X
- σY = Standard Deviation of Y

→ Correlation always lies between **-1 to 1**

- $+1 \rightarrow$ Perfect positive correlation
- $-1 \rightarrow$ Perfect negative correlation
- $0 \rightarrow$ No correlation

Correlation will mostly be used in Machine Learning which we will get to know later

Correlation

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Correlation

Let's understand correlation with an example,

Let's continue with our study hours and exam marks.

We found covariance = **23.75** (positive).

Standard deviation of hours (X) = 2.24

Standard deviation of marks (Y) ≈ 11.08

$$r = \frac{23.75}{2.24 \times 11.08} \approx 0.95$$

Hours Studied	Marks
2	50
4	60
6	65
8	80

So correlation = **0.95**,

which is very close to +1. ⟶ That means there's a **very strong positive relation.**